

SIOUX GATEWAY, IA, USA

WMO: 725570

Lat: 42.400N

Lon: 96.383W

Elev: 1102

StdP: 14.12

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-2.4	-13.8	2.7	-5.9	-9.7	3.4	-1.7	36.2	18.7	32.5	15.9	9.1	320	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	92.9	75.4	90.1	74.4	87.5	73.2	79.1	88.6	77.3	86.8	75.5	84.5	13.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.3	142.9	85.6	74.4	133.8	83.8	72.6	126.0	82.0	43.4	88.9	41.6	86.9	39.9	84.6	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.4	25.1	22.0	-15.3	97.9	4.7	3.4	-18.7	100.3	-21.4	102.3	-24.1	104.2	-27.5	106.6		
			-15.7	82.4	4.6	1.3	-19.0	83.3	-21.7	84.1	-24.3	84.8	-27.6	85.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	20.4	24.6	37.2	49.6	61.2	71.2	75.0	72.3	64.8	51.3	37.2	24.6
	DBStd	21.48	12.31	12.52	12.40	9.96	8.84	6.79	5.75	5.47	8.35	9.45	10.36	11.30
	HDD50	3482	919	711	426	124	11	0	0	0	4	98	400	789
	HDD65	6689	1384	1131	862	470	182	22	3	7	106	434	834	1254
	CDD50	3208	0	1	30	113	358	637	774	692	447	140	16	0
	CDD65	936	0	0	1	8	64	209	312	234	98	10	0	0
	CDH74	9706	0	0	27	175	774	2189	3224	2102	1056	154	5	0
	CDH80	3606	0	0	4	51	275	832	1317	726	359	42	0	0
	WSAvg	10.1	10.9	11.1	11.5	12.3	11.2	9.5	7.9	7.5	8.7	9.9	10.5	10.4
	PrecAvg	27.6	0.7	0.7	1.8	2.8	3.7	4.1	3.3	3.5	2.8	2.0	1.1	0.9
(15) Precipitation	PrecMax	41.5	2.7	2.7	5.9	6.7	7.5	16.7	10.3	10.1	9.7	6.9	4.6	2.9
	PrecMin	14.3	0.1	0.1	0.0	0.5	0.9	0.6	0.4	0.1	0.3	0.1	0.0	0.1
	PrecStd	5.6	0.5	0.6	1.3	1.6	1.5	2.5	2.2	2.2	1.9	1.6	1.0	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	61.5	77.7	86.3	92.9	95.2	97.3	94.7	93.0	86.1	72.4	55.9
		MCWB	44.6	51.0	57.8	63.3	68.9	75.0	78.2	77.1	72.3	67.1	57.1	47.6
	2%	DB	46.6	52.8	70.0	79.0	86.6	91.3	93.0	90.1	87.9	78.1	65.1	49.5
		MCWB	39.5	45.1	55.7	59.9	67.1	73.4	77.7	76.7	70.9	62.2	53.4	42.2
	5%	DB	40.8	46.2	62.2	73.1	82.5	88.1	90.0	86.9	84.2	72.9	59.0	43.9
		MCWB	36.0	40.5	52.8	57.2	64.8	71.9	76.6	74.8	69.8	59.8	49.2	38.4
	10%	DB	36.6	41.0	56.2	67.8	77.7	84.9	87.1	84.1	80.4	68.1	53.3	39.5
		MCWB	33.6	37.0	48.0	54.1	62.8	70.4	75.2	72.8	67.5	57.0	45.9	35.7
	0.4%	WB	44.5	52.3	62.4	65.7	73.1	78.9	82.1	80.4	76.2	70.1	59.7	50.0
		MCDB	54.0	59.8	71.9	80.0	86.0	90.0	91.1	90.2	86.3	78.2	67.8	54.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.9	45.3	57.7	62.0	70.1	76.7	80.0	78.1	74.0	65.5	54.9	42.5
		MCDB	45.5	52.3	65.4	75.2	81.7	87.5	89.1	87.4	84.0	73.6	64.2	48.3
	5%	WB	36.4	40.6	53.3	58.9	67.3	74.6	78.1	76.4	72.1	61.9	50.3	38.9
		MCDB	40.3	46.0	62.9	71.1	78.5	84.7	87.4	84.8	81.3	70.2	57.9	43.4
	10%	WB	33.6	37.1	48.0	55.3	64.5	72.6	76.5	74.5	69.7	58.2	46.1	35.9
		MCDB	36.4	40.7	56.6	65.4	75.0	82.1	85.6	82.0	77.6	66.8	53.0	39.4
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	18.5	20.8	23.4	22.5	21.4	20.8	20.7	24.4	23.7	21.0	17.6
		MCDBR	25.4	25.3	32.5	33.7	29.2	24.9	23.2	23.5	28.0	31.4	30.6	25.2
		MCWBR	19.1	18.4	19.7	17.5	13.5	11.4	11.4	12.0	13.6	16.9	19.2	18.7
	5% WB	MCDBR	23.6	24.0	30.0	28.4	24.5	21.0	20.6	20.6	23.0	25.6	28.1	23.1
		MCWBR	18.2	17.8	18.7	16.1	12.9	11.6	11.6	11.6	12.4	15.7	18.6	17.7
(40) Clear-Sky Solar Irradiance	taub		0.279	0.291	0.323	0.387	0.413	0.417	0.427	0.422	0.374	0.330	0.294	0.282
	taud		2.431	2.423	2.406	2.241	2.240	2.291	2.264	2.271	2.369	2.478	2.498	2.445
	Ebn at Noon		277	291	293	279	274	272	268	265	271	273	269	265
	Edh at Noon		24	29	33	42	43	41	42	40	34	27	23	22
(44) All-Sky Solar Radiation	RadAvg		614	953	1212	1550	1815	2027	2094	1760	1424	987	689	502
	RadStd		56	76	94	119	135	150	109	127	95	120	43	42

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page